ROADSENSE AI

ARJUN BAIJU

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Overview

- Introduction
- Exsisting System
- Proposed System
- Software and Hardware Requirements
- Problem Statement
- System Architecture
- Literature Survey
- System Design
- Implementation
- Result
- Reference

Introduction

- Traffic signs play a vital role in ensuring road safety.
- Drivers may miss traffic signs due to distractions or poor visibility.
- RoadSense AI uses the YOLO deep learning algorithm for real-time traffic sign detection.
- The system recognizes traffic signs and provides voice alerts to improve driver awareness.

Existing System

- Relies on manual observation by drivers.
- Traditional image processing methods have limited accuracy.
- Performance decreases under poor lighting and weather conditions.
- Detection of small or damaged traffic signs is difficult.

Proposed System

- Uses the YOLOv8 deep learning model for traffic sign detection.
- Processes live video captured through a webcam.
- Detects and classifies traffic signs in real time.
- Displays detected signs with bounding boxes and confidence scores.
- Generates voice alerts to assist the driver.

Software Requirements

- **Operating System:** Windows 11
- **Programming Language:** Python 3.14
- **Web Framework:** Flask
- **Deep Learning Model:** YOLOv8 (Ultralytics)
- **Libraries:** OpenCV, PyTorch, NumPy, Pyttsx3, MySQL Connector/Python
- **Database:** MySQL Community Server 8.0
- **Dataset:** Roboflow Traffic Sign Dataset
- **IDE:** Visual Studio Code
- **Frontend Technologies:** HTML5, CSS3, JavaScript, Bootstrap 5

Hardware Requirements

- **Processor:** Intel Core i5 (13th Generation or above)
- **RAM:** 16 GB
- **Graphics Card:** NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM)
- **Camera:** HD Webcam or Vehicle-Mounted Camera
- **Storage:** Minimum 20 GB Free Disk Space
- **Internet:** Required for dataset download and software updates

Problem Statement

*RoadSense AI **aims to develop** a real-time traffic sign detection and driver assistance system using YOLOv8 that detects traffic signs, provides voice alerts, and helps drivers make timely and safe decisions.*

System Architecture

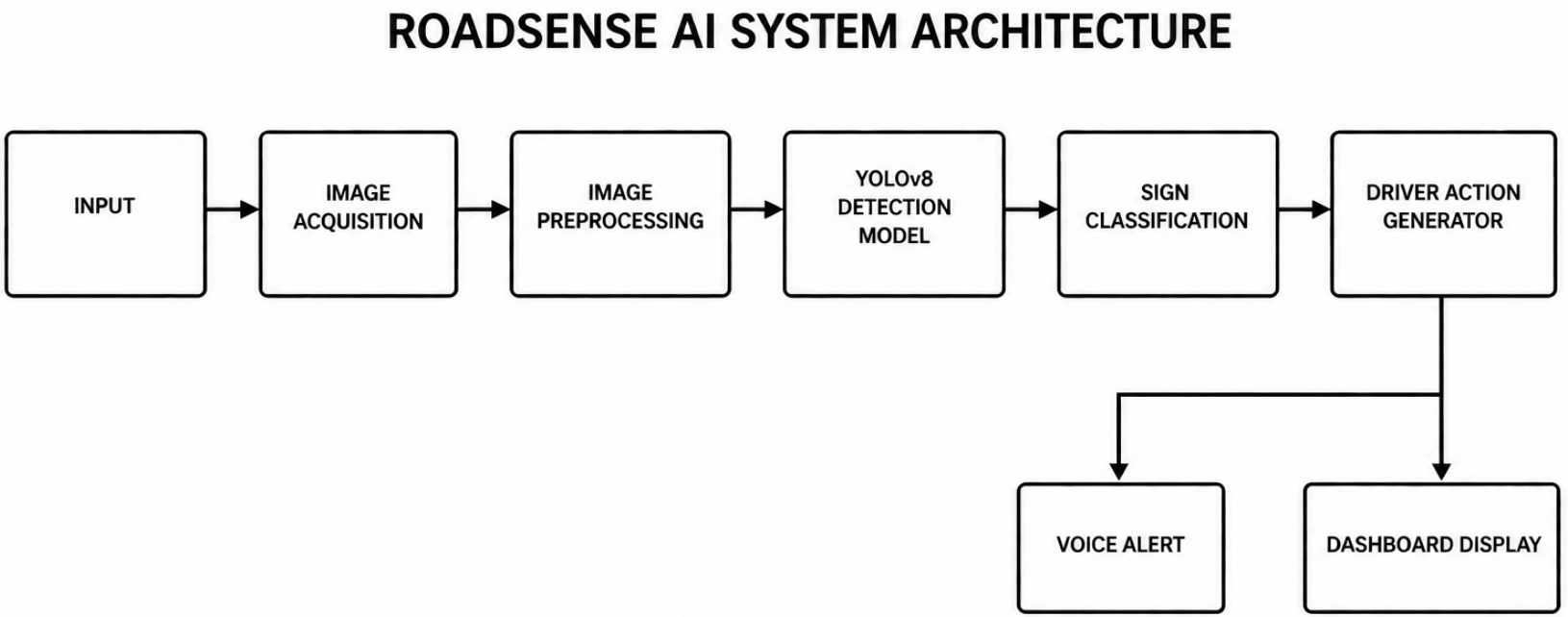


Figure: Architecture of the RoadSense AI System

Literature Survey (1/5)

Paper Title	Author(s)	Tools Used	Advantages	Disadvantage
A cascaded R-CNN with multiscale attention and imbalanced samples for traffic sign detection [1]	Zhang, Z. Xie, J. Sun, X. Zou, and J. Wang (2020)	Cascade R-CNN, Multiscale Attention, FPN, CNN	Improves detection of small traffic signs; handles imbalanced samples; high detection accuracy.	Complex architecture; high computational cost; slower inference.
Integrated feature pyramid network with feature aggregation for traffic sign detection [2]	Q. Tang, G. Cao, and K.-H. Jo (2021)	IFA-FPN, Feature Aggregation, Cascade R-CNN, FPN	Better feature representation; improved small sign detection; higher mAP.	High computational complexity; requires quality datasets.

Literature Survey (2/5)

Paper Title	Author(s)	Tools Used	Advantages	Disadvantage
An improved faster R-CNN for small object detection [3]	C. Cao, B. Wang, W. Zhang, X. Zeng, X. Yan, Z. Feng, Y. Liu, and Z. Wu (2020)	Faster R-CNN, CNN, Region Proposal Network (RPN), Feature Pyramid	Improves localization and detection of small objects; higher detection accuracy.	Slow inference speed; computationally expensive.
Traffic sign detection using feature fusion and contextual information [4]	H. Wang, G. Chen, Z. Li, and Z. Liu (2021)	Feature Fusion, Contextual Information, CNN	Improves robustness and detection accuracy in complex scenes.	Higher computation; depends on scene context.

Literature Survey (3/5)

Paper Title	Author(s)	Tools Used	Advantages	Disadvantage
Traffic sign recognition algorithm based on feature pyramid attention [5]	P. Su, W. Li, L. Sha, Y. Shi, and T. Dong (2021)	Feature Pyramid Attention, CNN, Deep Learning	Improves recognition accuracy; enhances feature extraction.	Focuses mainly on recognition; increased computational cost.
Traffic sign detection and recognition using multi-scale fusion and prime sample attention [6]	J. Cao, J. Zhang, and W. Huang (2021)	Multi-scale Fusion, Prime Sample Attention, CNN	Improves detection of different traffic sign sizes.	Requires more GPU memory and training time.

Literature Survey (4/5)

Paper Title	Author(s)	Tools Used	Advantages	Disadvantage
Improved faster R-CNN traffic sign detection based on a second region of interest and highly possible regions proposal network [7]	F. Shao, X. Wang, F. Meng, J. Zhu, D. Wang, and J. Dai (2020)	Faster R-CNN, Second RoI, Region Proposal Network (RPN), CNN	Better localization accuracy; reduces false detections.	High computational cost; slower than one-stage detectors.
Improved traffic sign detection algorithm based on faster R-CNN [8]	X. Gao, L. Chen, K. Wang, X. Xiong, H. Wang, and Y. Li (2022)	Faster R-CNN, CNN, Feature Extraction	Improves robustness and detection accuracy.	Not suitable for low-power embedded devices.

Literature Survey (5/5)

Paper Title	Author(s)	Tools Used	Advantages	Disadvantage
Improved faster R-CNN traffic sign detection algorithm based on transformer [9]	C. Tian, Z. Liu, H. Liu, Y. Yan, and J. Zhang (2023)	Transformer, Faster R-CNN, CNN, PyTorch	Improves detection of small and occluded traffic signs.	Requires powerful GPU; high compu-tational complexity.
Traffic sign detection method based on improved SSD [10]	You, Q. Bi, Y. Ji, S. Liu, Y. Feng, and F. Wu (2020)	Improved SSD, CNN, Feature Fu-sion	Improves SSD accu-racy while maintaining real-time perfor-mance.	Lower ac-curacy than recent YOLO-based meth-ods.

Conclusion of Literature Survey

- Various deep learning methods have been proposed for traffic sign detection.
- The surveyed techniques include **Faster R-CNN, Cascade R-CNN, SSD, FPN,** and **Transformer-based models**.
- These methods provide high detection accuracy, especially for small traffic signs.
- However, they require high computational resources and have slower processing speed.
- Based on the literature survey, **YOLO** is selected for this project.
- **YOLO** offers a good balance of accuracy, speed, and real-time performance, making it suitable for traffic sign detection.

BLOCK DIAGRAM

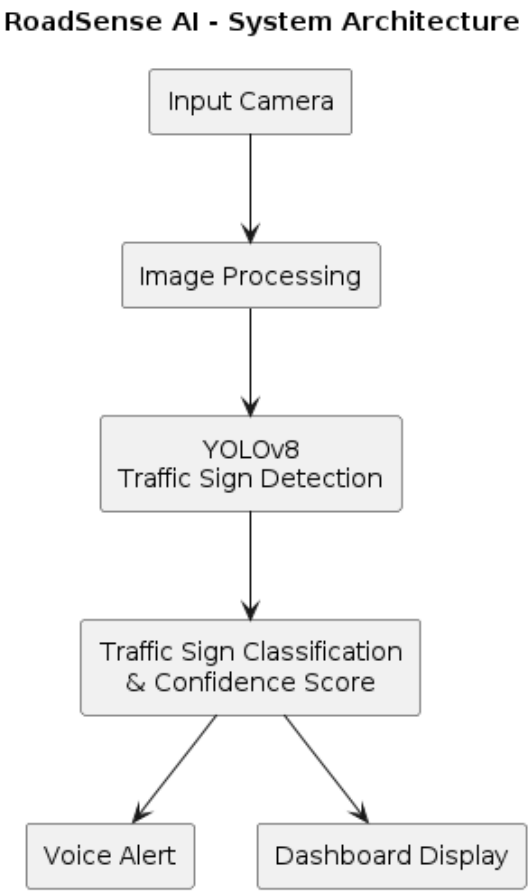


Figure: Block diagram of the RoadSense AI System

USE CASE DIAGRAM

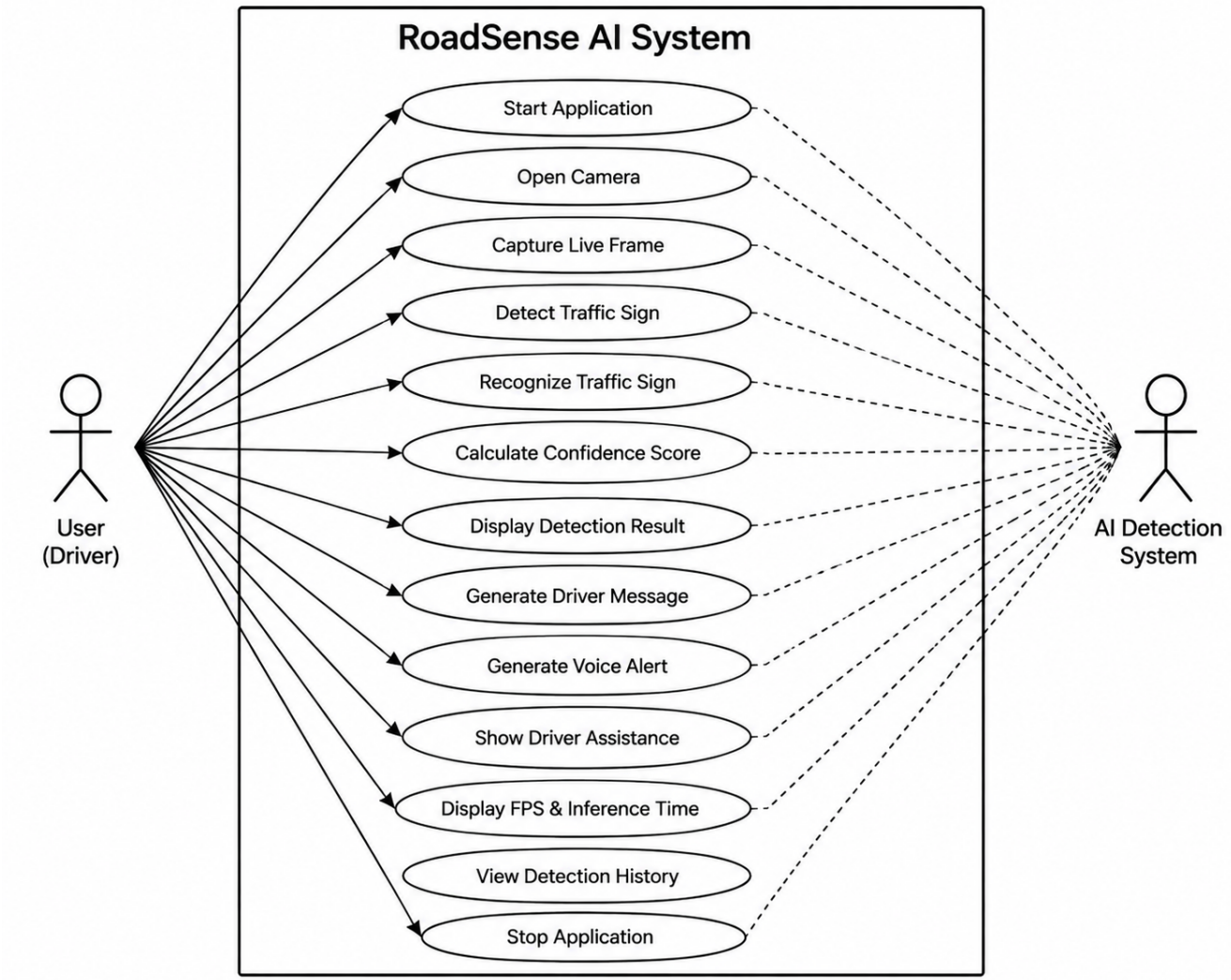


Figure: Use case diagram of the RoadSense AI System

SEQUENCE DIAGRAM FOR Driver/User

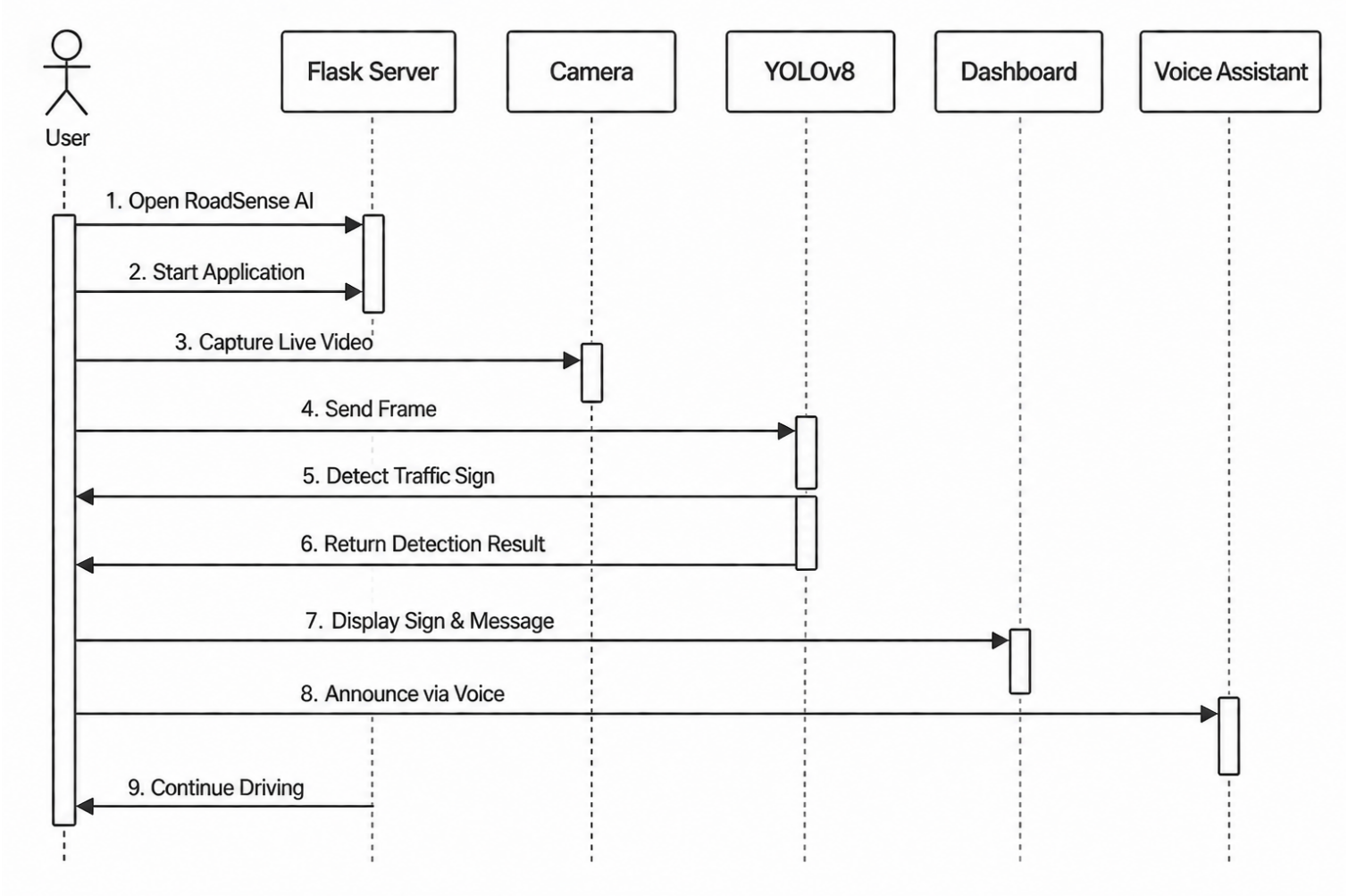


Figure: Sequence diagram for Driver/User for RoadSense Ai System

SEQUENCE DIAGRAM FOR Traffic sign Detection

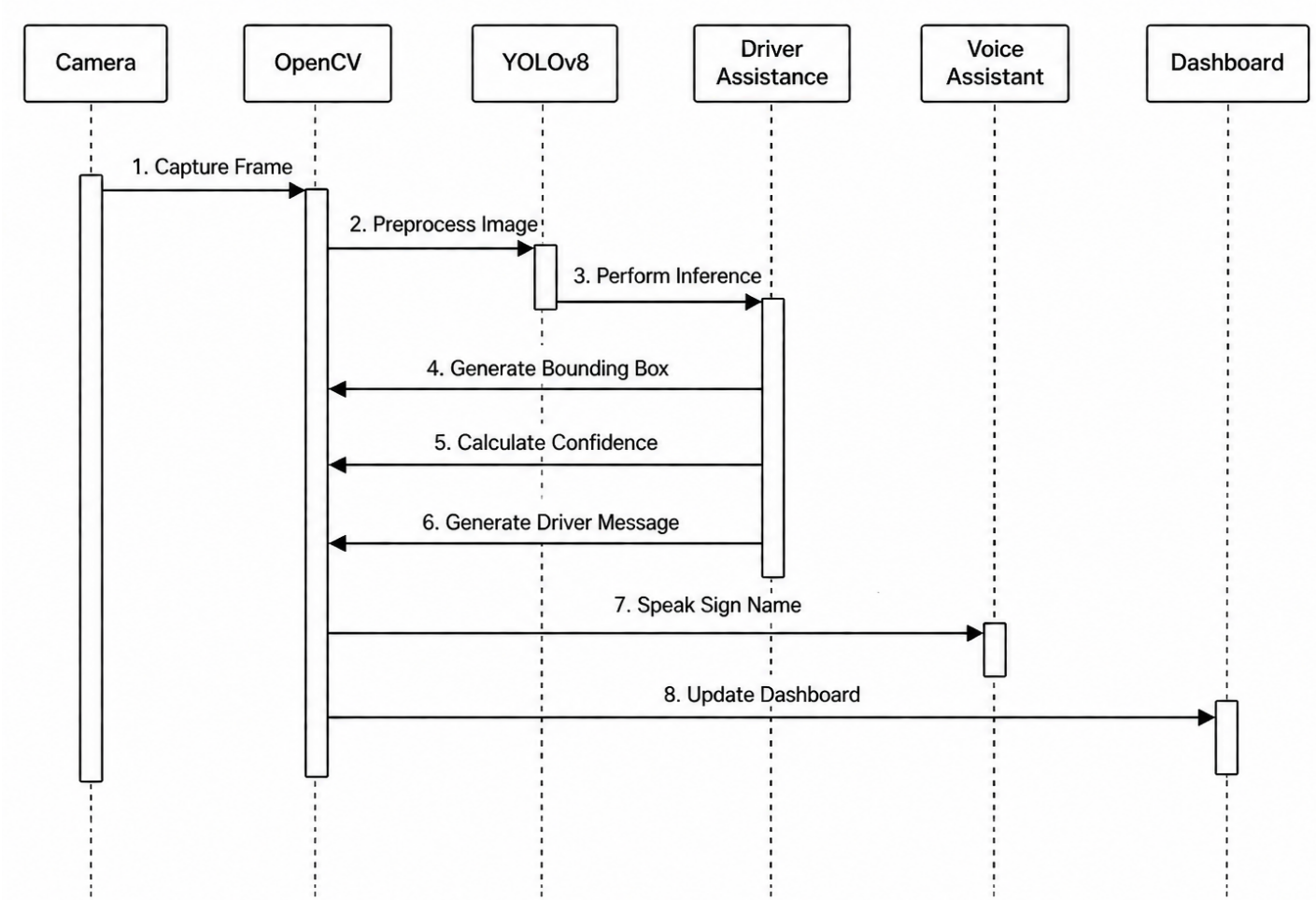


Figure: Sequence diagram of traffic Sign detection for RoadSense Ai

DFD LEVEL-0

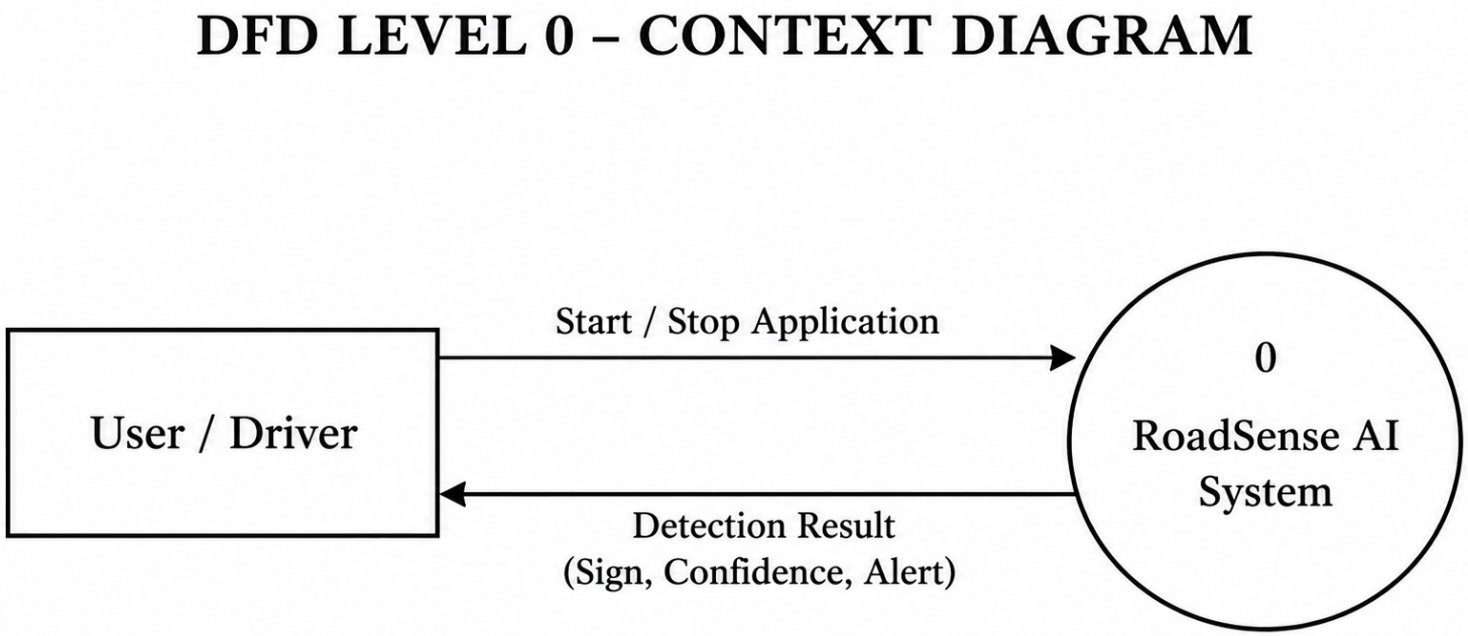


Figure: Dfd level-0 diagram for RoadSense Ai System

DFD LEVEL-1

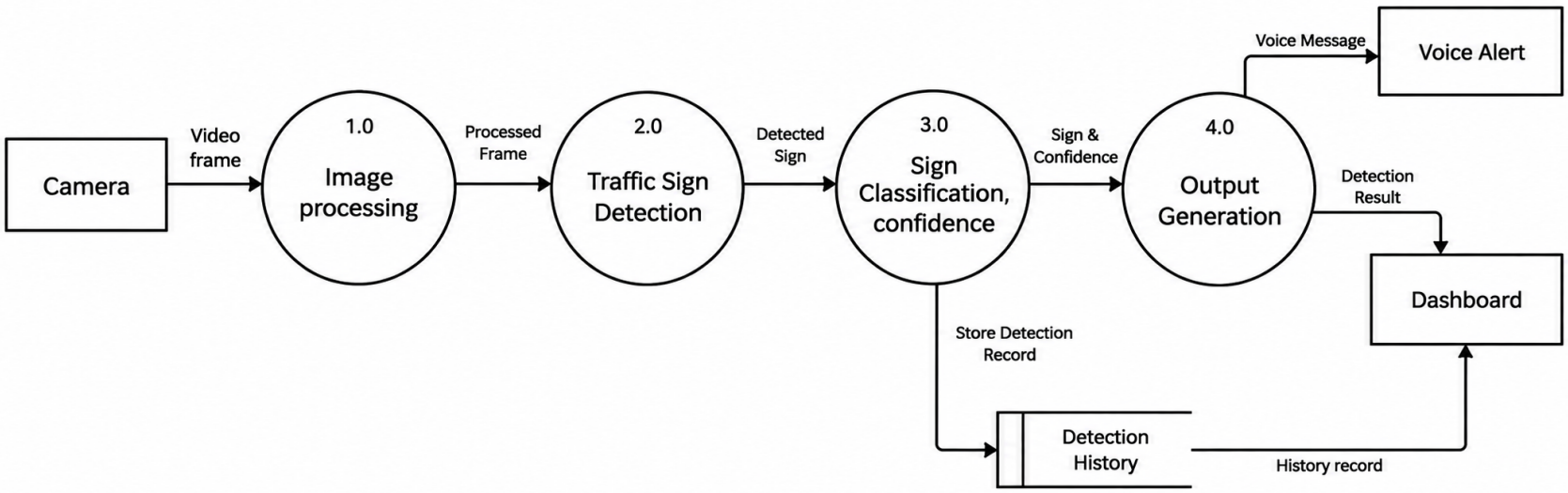


Figure: Dfd level-1 diagram for RoadSense Ai System

DFD LEVEL-2

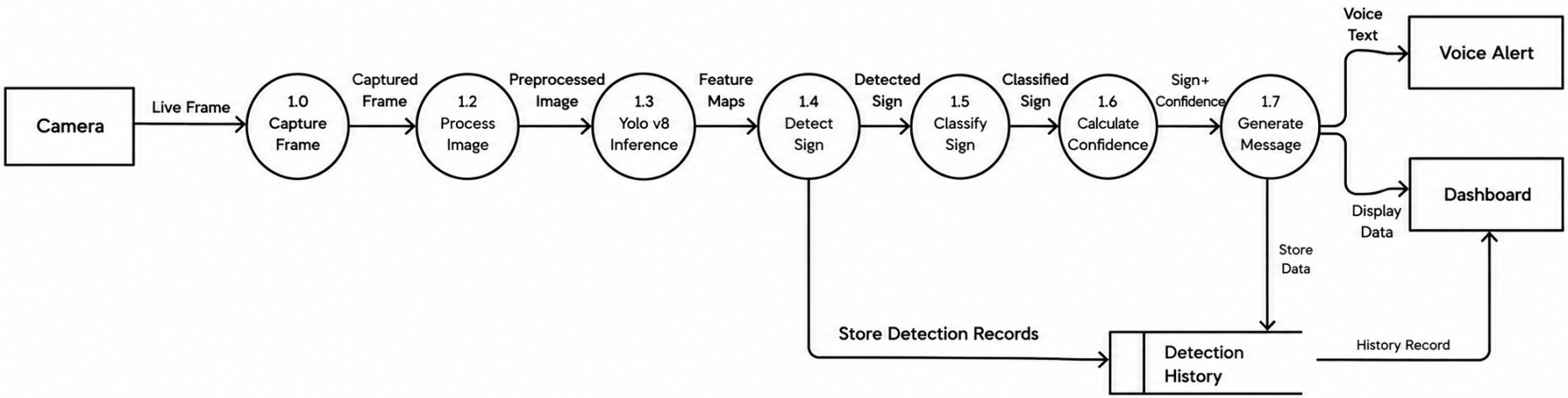


Figure: Dfd level-2 diagram for RoadSense Ai System

ER DIAGRAM

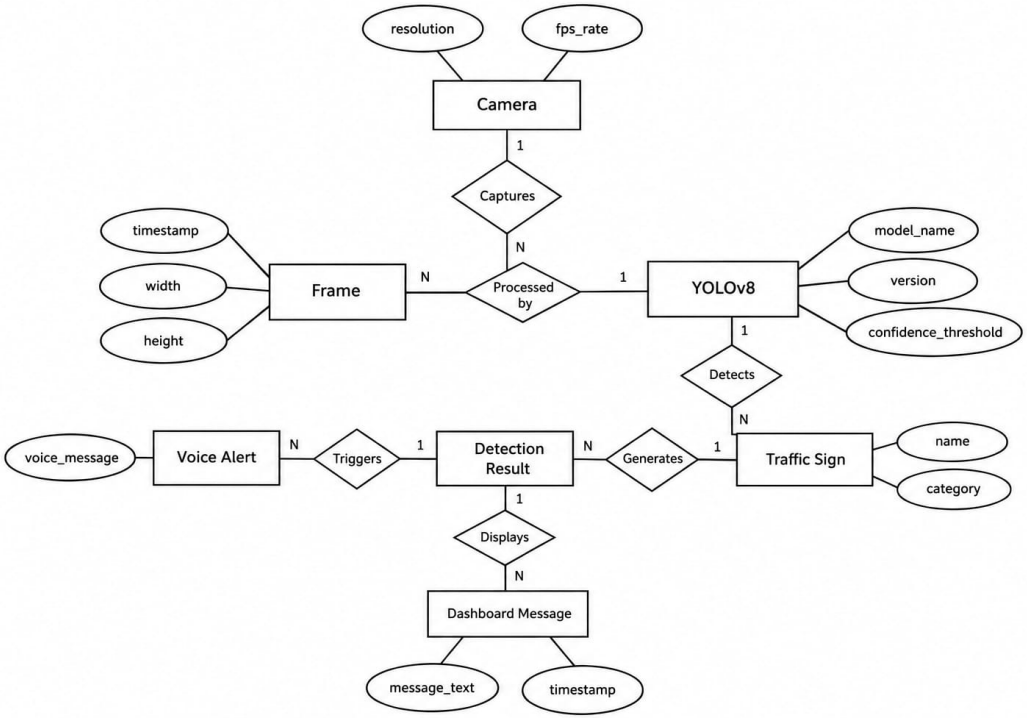


Figure: Er diagram for RoadSense Ai System

Gantt Chart

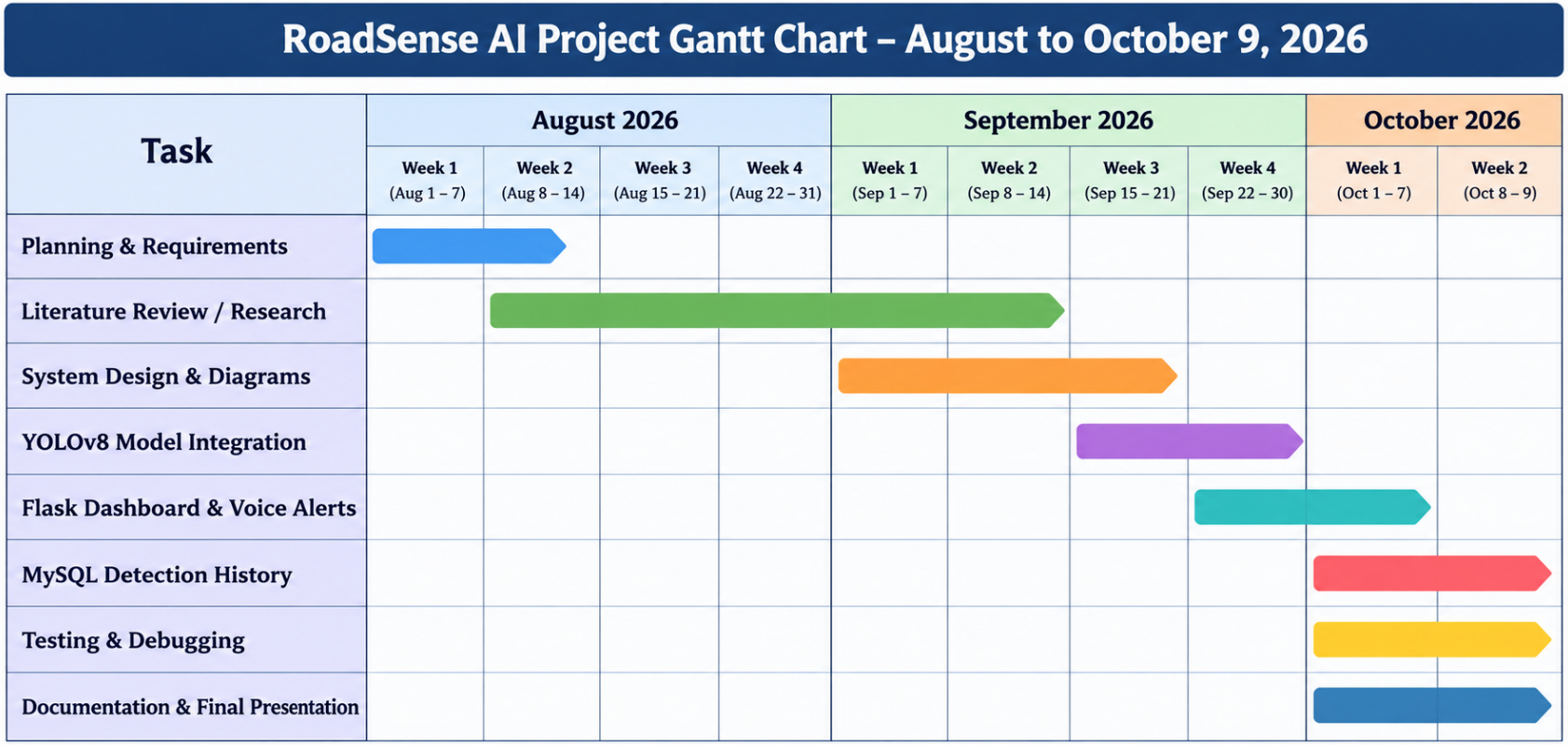


Figure: Gantt Chart of RoadSense Ai System

Implementation

The implementation phase converts the proposed design into a working AI-based traffic sign detection system:

- **Front-End Development:**Flask and Bootstrap dashboard for detection results and performance.
- **Image Processing:**Capture and preprocess live video using OpenCV.
- **YOLOv8 Integration:**Detect traffic signs and calculate confidence scores
- **Voice Alert System:**Generate driver alerts using Pyttsx3.
- **Database Management:**Store detection records in MySQL with timestamps.
- **Detection History:**Display real-time results and previous detection logs

interface

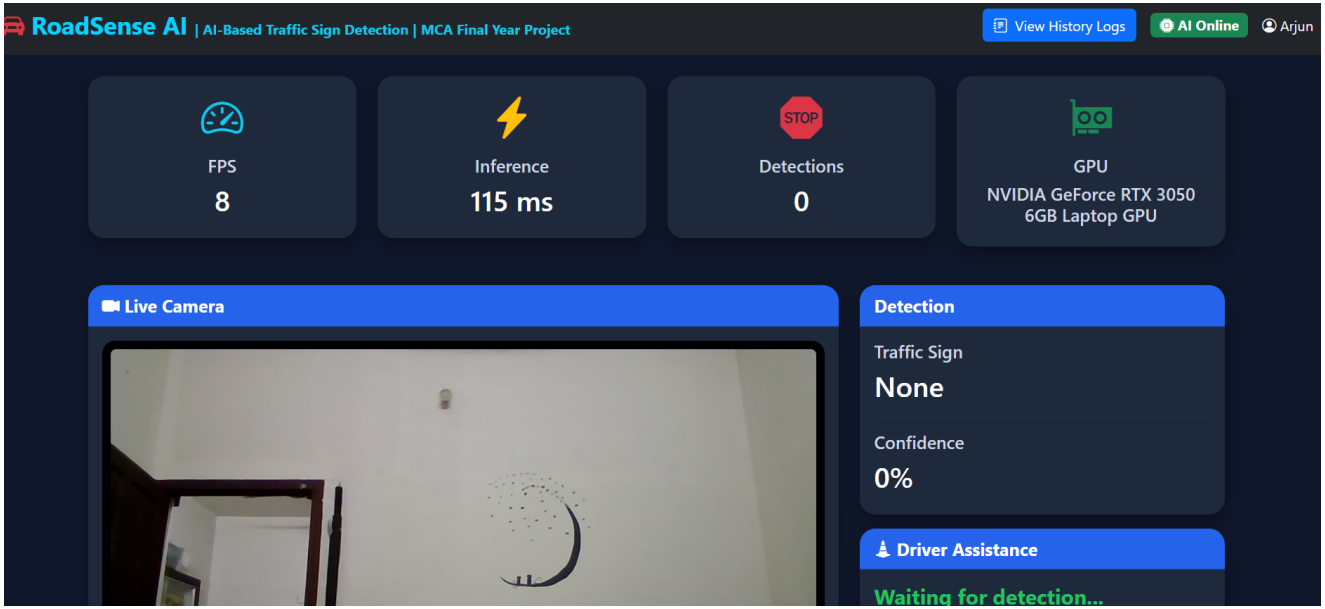


Figure: Interface of RoadSense AI System

Detection

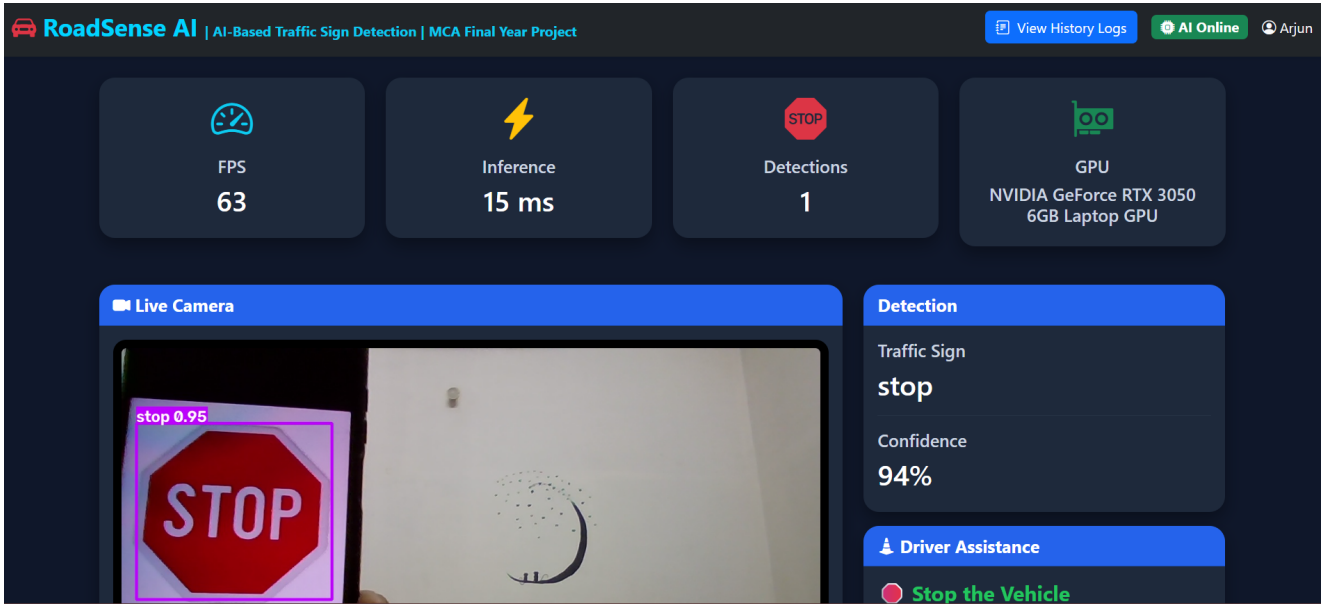


Figure: Detect Sign in the RoadSense AI System

Processing Sign

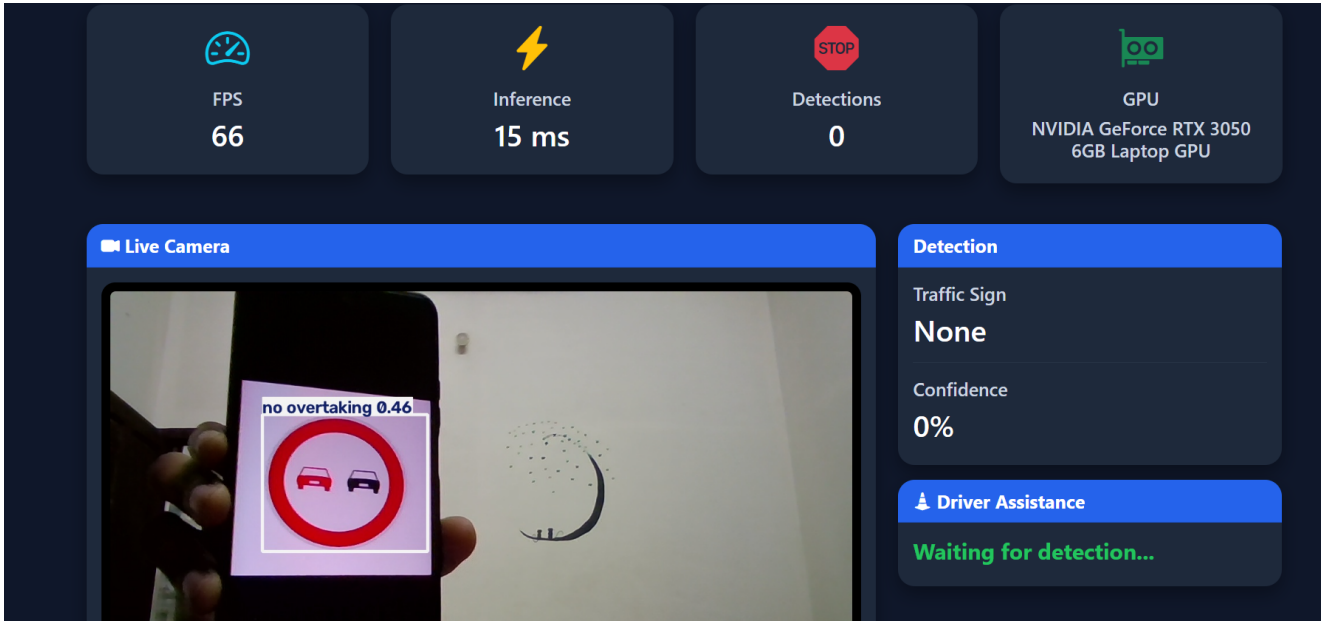


Figure: Process Sign in the RoadSense AI System

History

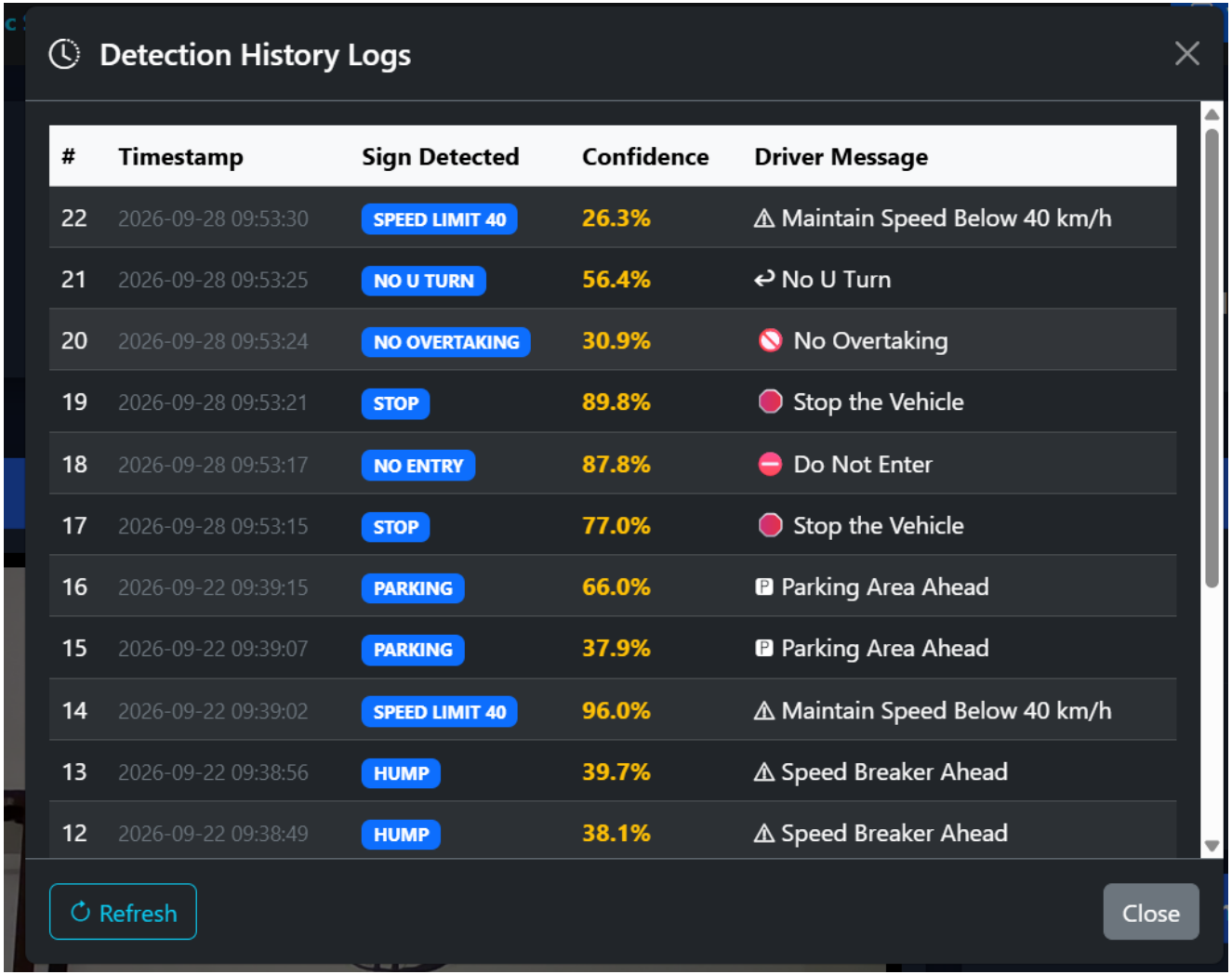


Figure: Stores the history of signs in RoadSense AI System

Output

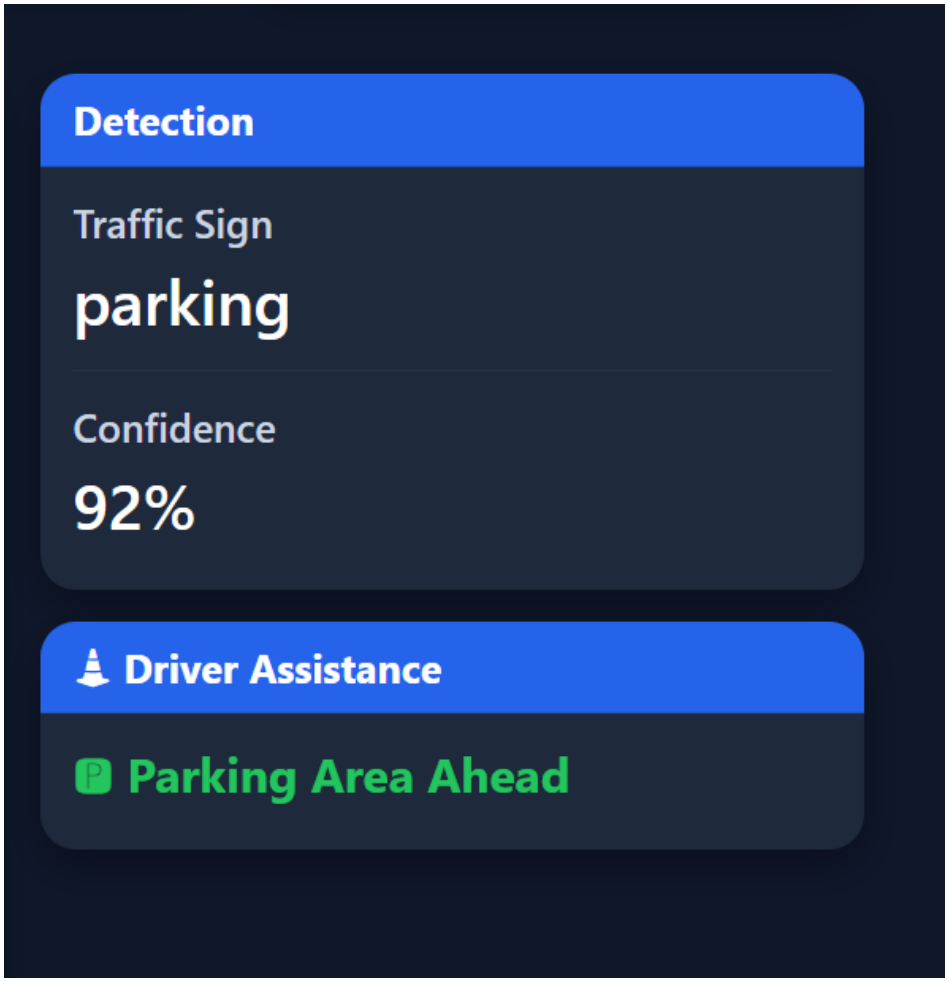


Figure: Output of the RoadSense AI System

Reference

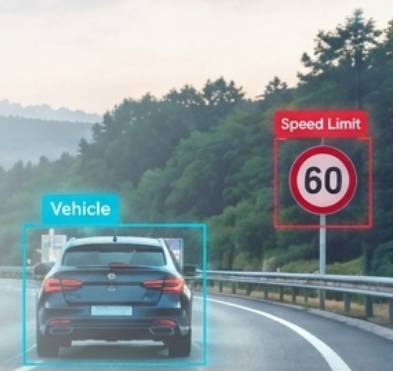
- *[1] Z. Zhang et al., "A cascaded R-CNN with multiscale attention and imbalanced samples for traffic sign detection," IEEE Access, vol. 8, pp. 29742–29754, 2020.*
- *[2] Q. Tang, G. Cao, and K.-H. Jo, "Integrated feature pyramid network with feature aggregation for traffic sign detection," IEEE Access, vol. 9, pp. 117784–117794, 2021.*
- *[3] C. Cao et al., "An improved faster R-CNN for small object detection," IEEE Access, vol. 7, pp. 106838–106846, 2020.*
- *[4] H. Wang et al., "Traffic sign detection using feature fusion and contextual information," in Proc. IEEE ICARM, pp. 949–953, 2021.*
- *[5] P. Su et al., "Traffic sign recognition algorithm based on feature pyramid attention," Journal of Physics: Conference Series, vol. 2035, no. 1, 2021.*

RoadSense AI

Smarter Vision. Safer Roads.

Thank You

for your attention



Detect Safer

Understand Better

Analyze Performance